Topic 1: Bullous Pemphigoid vs Pemphigus Vulgaris

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Autoimmune blistering diseases. Bullous Pemphigoid (BP): IgG autoantibodies against hemidesmosomes (BP180/BP230), causing subepidermal tense bullae. Pemphigus Vulgaris (PV): IgG against desmoglein 1/3, intraepidermal flaccid bullae.

CLINICAL PRESENTATION & PHYSICAL EXAM

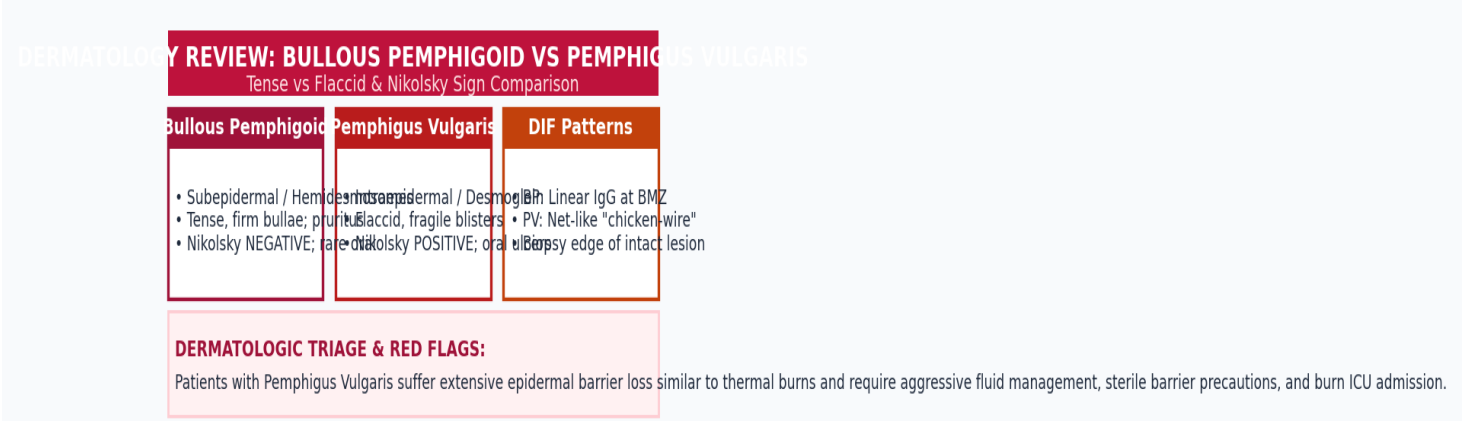
• **Presentation:** BP: elderly patient, tense blisters on urticarial base, severe pruritus, mucous membranes rarely involved. PV: painful flaccid blisters, oral mucosal erosions precede skin lesions, easily ruptured.
• **Physical Exam:** BP: Negative Nikolsky sign (rubbing normal skin does not induce blister), tense bullae. PV: Positive Nikolsky sign, positive Asboe-Hansen sign (lateral blister extension).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Skin biopsy with Direct Immunofluorescence (DIF). BP: linear IgG and C3 deposition along the basement membrane zone. PV: "chicken-wire" or "net-like" IgG pattern in epidermis.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

High-dose systemic Corticosteroids (Prednisone 0.5-1 mg/kg/day) or topical superpotent steroids (Clobetasol). Immunosuppressive steroid-sparing agents (Azathioprine, Mycophenolate). Wound care, secondary infection prevention.



HIGH-YIELD BOARD PEARL: Bullous Pemphigoid features TENSE bullae and a NEGATIVE Nikolsky sign without prominent mucosal involvement. Pemphigus Vulgaris features FLACCID bullae, oral ulcers, and a POSITIVE Nikolsky sign.

Topic 2: Thermal Burns & Resuscitation

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Tissue injury from thermal energy. Depth: Superficial (1st degree), Partial-thickness (2nd degree: blistering, painful), Full-thickness (3rd degree: leathery, painless, thrombosed vessels), 4th degree (bone/muscle).

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Severe pain, erythema, blisters, or painless charred skin. Inhalation injury signs: singed facial hair, soot in airway, stridor, hoarseness, carbonaceous sputum.
• **Physical Exam:** Rule of Nines for % TBSA: Head 9%, Each Arm 9%, Chest 9%, Abdomen 9%, Upper Back 9%, Lower Back 9%, Each Leg 18%, Perineum 1%. Partial/full thickness count toward TBSA.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

ABG with Co-oximetry (Carboxyhemoglobin level for Carbon Monoxide, Cyanide level if synthetic materials burned). CXR, bronchoscopy for inhalation injury. CBC, electrolytes, CK for rhabdomyolysis.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Parkland Formula: 4 mL x weight (kg) x % TBSA (2nd/3rd degree burns). Give 50% in first 8 hours (from time of burn), remaining 50% over next 16 hours using Lactated Ringer. Titrate to urine output 0.5-1 mL/kg/h. Early endotracheal intubation for inhalation airway edema. Escharotomy for circumferential burns.



HIGH-YIELD BOARD PEARL: Parkland formula fluid is calculated from the TIME OF INJURY, not time of ED arrival. Target urine output is 0.5 to 1 mL/kg/hour in adults. High-risk inhalation signs mandate prophylactic early intubation before airway edema closes the glottis.

Topic 3: Cellulitis & Purulent Soft Tissue Infection

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Infection of the deep dermis and subcutaneous fat. Non-purulent: Streptococcus pyogenes (Group A Strep) predominantly. Purulent: Staphylococcus aureus (including MRSA).

CLINICAL PRESENTATION & PHYSICAL EXAM

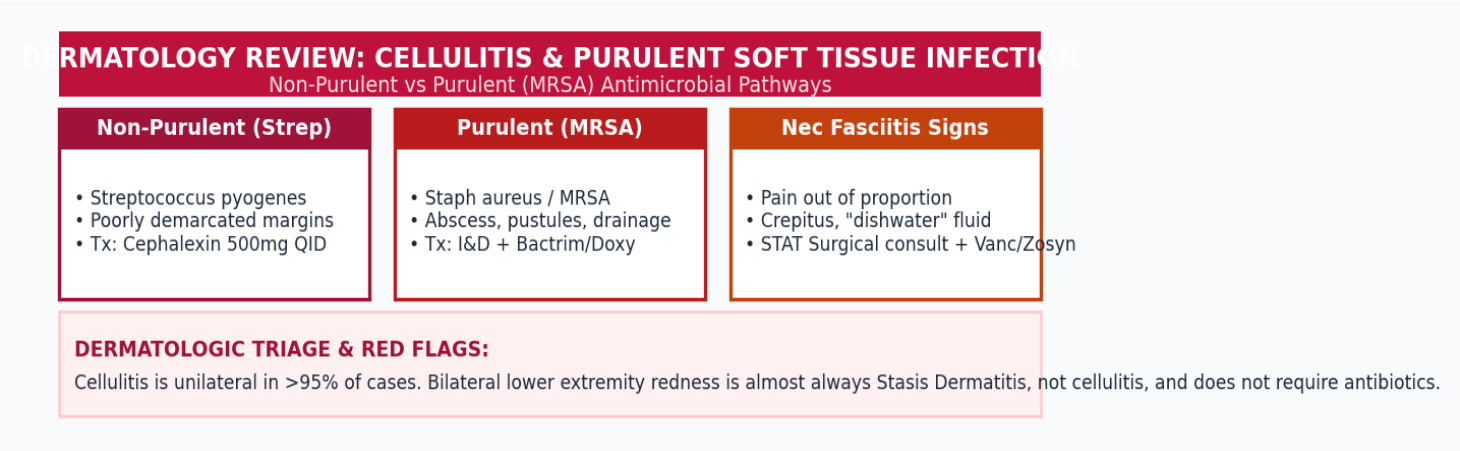
• **Presentation:** Spreading erythema, warmth, localized swelling, pain. Possible systemic symptoms (fever, chills, tachycardia).
• **Physical Exam:** Poorly demarcated, expanding border of erythema and edema. Regional lymphadenopathy. Fluctuance indicates underlying abscess requiring incision and drainage.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Primarily clinical. Demarcate borders with surgical pen to track progression. Bedside POCUS to detect occult abscess (fluid collection / "cobblestoning"). Blood cultures only if immunocompromised or systemic toxicity.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Non-purulent (Strep): Cephalexin 500 mg QID OR Amoxicillin-clavulanate. Purulent / MRSA risk: TMP-SMX DS 1-2 tabs BID OR Doxycycline 100 mg BID OR Clindamycin. Inpatient IV: Cefazolin (Strep/MSSA) or Vancomycin (MRSA). Limb elevation.



HIGH-YIELD BOARD PEARL: Mark the advancing border of erythema with a pen to monitor treatment response. Pain out of proportion to exam, crepitus, skin necrosis, or hemorrhagic bullae are red flags for Necrotizing Fasciitis requiring emergent surgical debridement.

Topic 4: Contact Dermatitis & Atopic Eczema

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Contact Dermatitis: Irritant (non-immunologic physical/chemical injury, 80%) vs Allergic (Type IV delayed hypersensitivity: Urushiol/poison ivy, nickel, neomycin). Atopic Dermatitis: impaired epidermal barrier (filaggrin mutation) + IgE dysregulation.

CLINICAL PRESENTATION & PHYSICAL EXAM

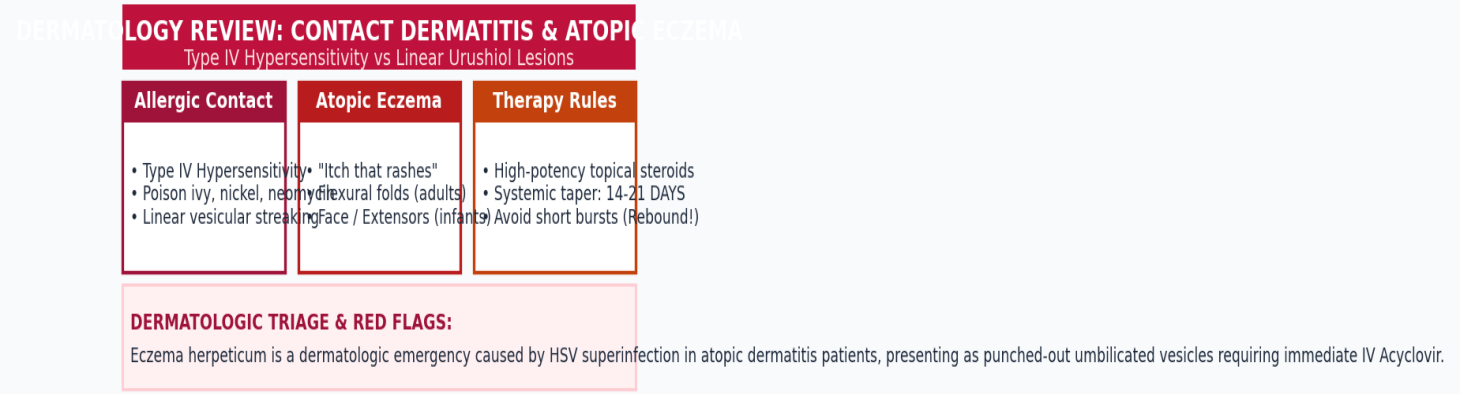
• **Presentation:** Allergic contact: intense pruritus, linear erythematous papules and vesicles (rhus dermatitis). Atopic: chronic pruritic, scaly, lichenified plaques in flexural folds (antecubital, popliteal; face/extensor in infants).
• **Physical Exam:** Erythematous vesicular rash with sharp linear borders matching contact exposure. Lichenification and excoriations in chronic eczema.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. Patch testing for allergic contact dermatitis outpatient.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Remove offending agent, wash skin immediately (Technu / dish soap for poison ivy). Topical high-potency Corticosteroids (Clobetasol, Triamcinolone; avoid face/groin - use low potency Hydrocortisone). Oral prednisone taper (14-21 days) for severe/extensive poison ivy. Oral antihistamines for pruritus.



HIGH-YIELD BOARD PEARL: Poison ivy (urushiol) dermatitis requires a prolonged oral prednisone taper of 14 to 21 days; shorter 5-day courses commonly result in severe rebound dermatitis.

Topic 5: Drug Eruptions (Exanthematous & Urticarial)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Adverse cutaneous drug reactions. Morbilliform/Exanthematous: Type IV T-cell mediated (most common, 90%), starts 7-14 days after starting drug (antibiotics, anticonvulsants, allopurinol). DRESS: Drug Reaction with Eosinophilia and Systemic Symptoms.

CLINICAL PRESENTATION & PHYSICAL EXAM

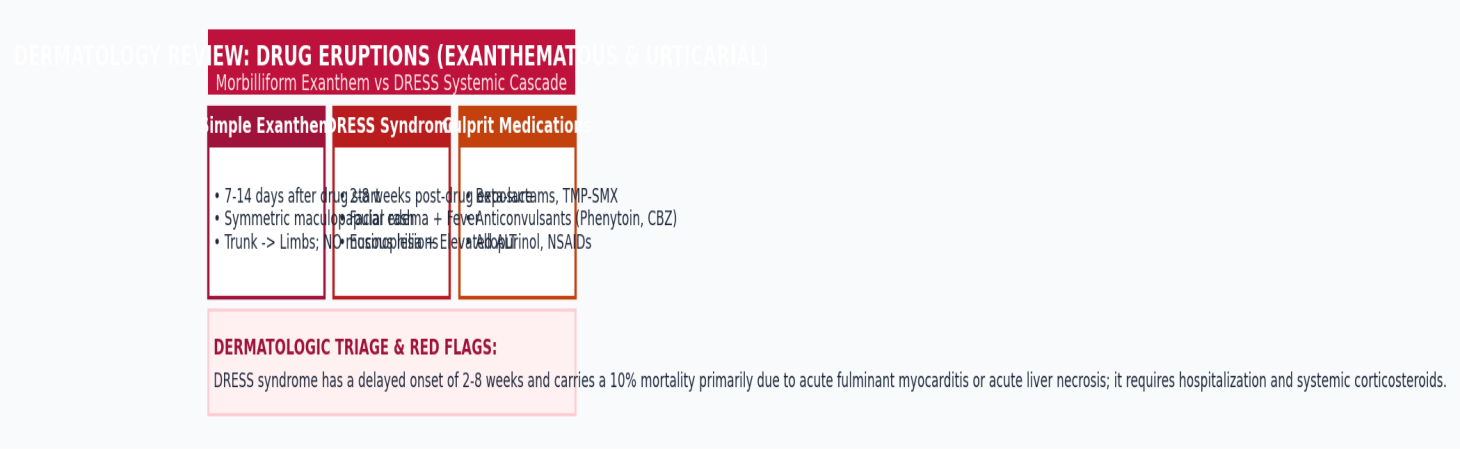
• **Presentation:** Symmetric, widespread maculopapular rash spreading centrifugally from trunk to extremities. Mild pruritus. Absence of systemic toxicity (fever, facial edema, lymphadenopathy favor DRESS).
• **Physical Exam:** Confluent pink-to-red macules and papules ("measles-like"). Sparing of mucous membranes. Normal vital signs.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. CBC with differential (check for eosinophilia > 1500 in DRESS). LFTs and creatinine to rule out systemic end-organ damage.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Discontinue offending medication immediately. Supportive care: oral H1 antihistamines (Cetirizine, Diphenhydramine), topical corticosteroids (Triamcinolone 0.1%), cool compresses. Systemic steroids if severe.



HIGH-YIELD BOARD PEARL: Morbilliform drug rashes typically appear 7-14 days after initiating the causative drug (most commonly penicillins, cephalosporins, or sulfa drugs) and spare mucous membranes.

Topic 6: Erysipelas

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Superficial skin infection involving the upper dermis and superficial cutaneous lymphatics. Pathogen: Streptococcus pyogenes (Group A Beta-hemolytic Strep) in the vast majority.

CLINICAL PRESENTATION & PHYSICAL EXAM

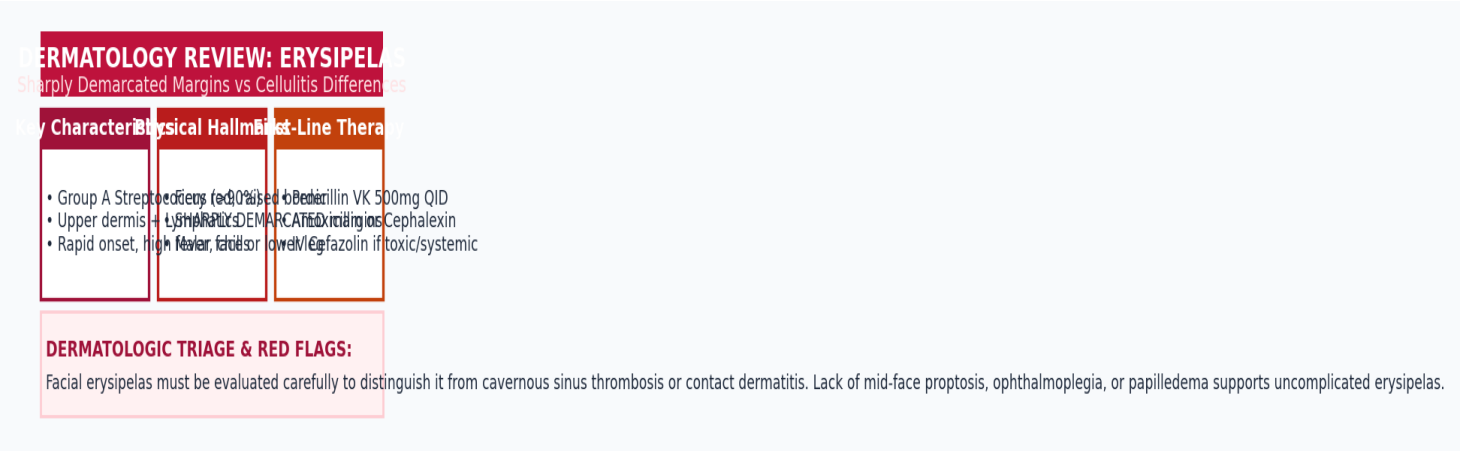
• **Presentation:** Sudden onset of fiery-red, painful, hot skin rash, typically on the lower extremities or face (malar distribution), accompanied by high fever, chills, and malaise.
• **Physical Exam:** Intensely erythematous, indurated plaque with SHARPLY DEMARCATED, RAISED BORDERS. "Peau d orange" (orange peel) texture due to lymphatic obstruction.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. Elevated ESR/CRP and leukocytosis with left shift.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Oral therapy for mild/stable: Penicillin VK 500 mg QID OR Amoxicillin OR Cephalexin. Penicillin-allergic: Clindamycin. Inpatient/Toxic: IV Penicillin G OR Cefazolin. Elevation of affected extremity.



HIGH-YIELD BOARD PEARL: Erysipelas is distinguished from cellulitis by its SHARPLY DEMARCATED, RAISED BORDERS and rapid involvement of superficial lymphatics with early high fevers; Group A Strep is the causative agent in >90%.

Topic 7: Herpes Zoster (Shingles) & Zoster Ophthalmicus

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Reactivation of latent Varicella-Zoster Virus (VZV) in the dorsal root ganglia or cranial nerve sensory ganglia (e.g., trigeminal ganglion).

CLINICAL PRESENTATION & PHYSICAL EXAM

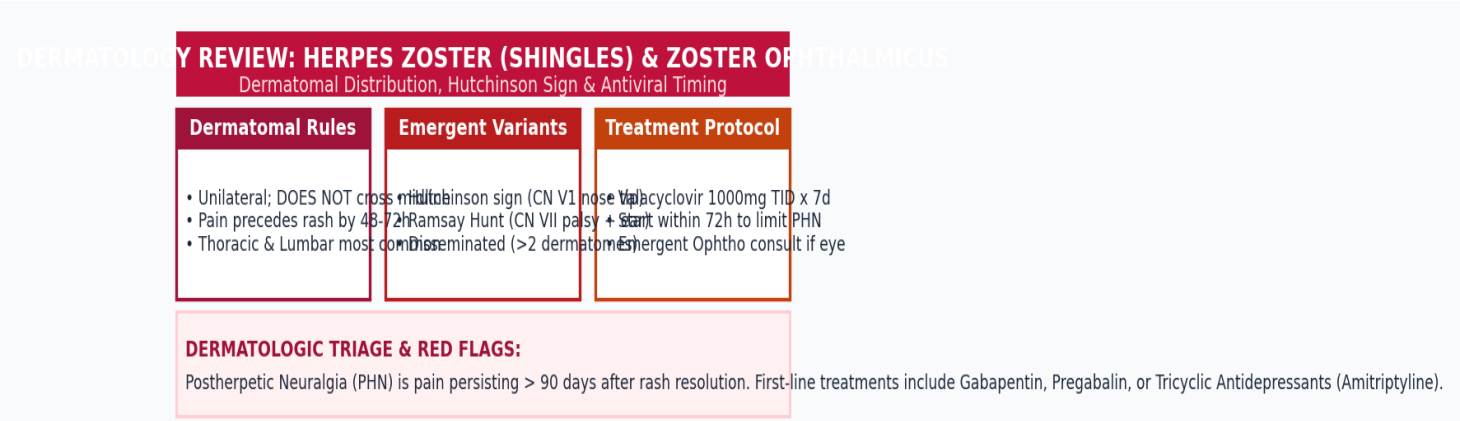
• **Presentation:** Pain, burning, or tingling prodrome followed by a painful eruption of grouped vesicles on an erythematous base in a unilateral, single dermatomal distribution.
• **Physical Exam:** Grouped vesicular rash that DOES NOT CROSS THE MIDLINE. Hutchinson sign: vesicle on tip of nose (involves nasociliary branch of CN V1; high risk of ocular corneal blindness). Ramsay Hunt: facial palsy + ear canal vesicles (CN VII).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical. Fluorescein slit-lamp exam for all V1 zoster (dendritic corneal lesions). PCR of vesicular fluid if atypical.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Valacyclovir 1000 mg TID x 7d OR Acyclovir 800 mg 5 times/day x 7d (start within 72h of rash onset). Hutchinson sign / Zoster Ophthalmicus: STAT Ophthalmology consult + IV/Oral antiviral. Corticosteroids for Ramsay Hunt.



HIGH-YIELD BOARD PEARL: Hutchinson sign (vesicles on the tip, side, or root of the nose) indicates nasociliary nerve involvement of CN V1 and mandates immediate slit-lamp fluorescein examination and ophthalmology consultation for Zoster Ophthalmicus.

Topic 8: Impetigo (Non-Bullous & Bullous)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Superficial bacterial skin infection of epidermis. Non-bullous (70%): Staphylococcus aureus or Group A Streptococcus. Bullous (30%): Staph aureus strains producing exfoliative toxin A (targets desmoglein 1).

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Child with facial lesions around nose/mouth. Honey-colored crusts (non-bullous) or thin-walled flaccid bullae that rupture leaving collarette of scale.

• **Physical Exam:** Erythematous macules/papules that develop vesicles and rupture, forming characteristic HONEY-COLORED CRUSTS. Usually painless; regional lymphadenopathy.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. Bacterial culture of lesion base if refractory or epidemic outbreak.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Localized (limited lesions): Topical Mupirocin 2% ointment TID x 5-7 days OR Retapamulin. Widespread / Bullous / Systemic: Oral Cephalexin 25-50 mg/kg/day OR Amoxicillin-clavulanate. MRSA suspected: Clindamycin or TMP-SMX.



HIGH-YIELD BOARD PEARL: Honey-colored golden crusts on the face or extremities are pathognomonic for non-bullous impetigo. Localized lesions are effectively treated with topical Mupirocin ointment alone.

Topic 9: Pediculosis (Lice: Capitis, Corporis, Pubis)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Infestation by blood-sucking lice. Head (Pediculus humanus capitis), Body (Pediculus humanus corporis - vector for epidemic typhus, trench fever, relapsing fever), Pubic ("Crabs", Pthirus pubis - sexually transmitted).

CLINICAL PRESENTATION & PHYSICAL EXAM

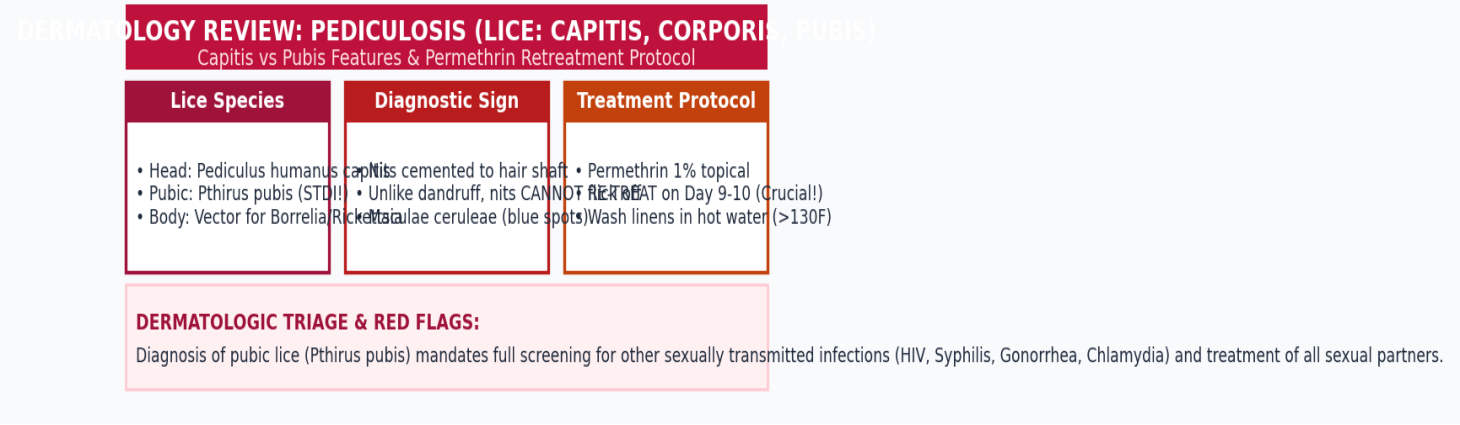
• **Presentation:** Intense scalp, body, or groin pruritus. Excoriations, secondary bacterial infection. Blue-gray macules (maculae ceruleae) on thighs/abdomen in pubic lice.
• **Physical Exam:** Visible live lice or nits (oval white eggs) firmly cemented to hair shafts within 6 mm of scalp. Cannot be dislodged easily like dandruff.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Direct visual inspection with fine-toothed comb and magnifying lens / Wood lamp.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

First-line: Topical Permethrin 1% lotion applied to affected areas, leave on for 10 min, rinse, and REPEAT IN 9-10 DAYS (kills newly hatched nymphs). Malathion 0.5% or Ivermectin for resistant cases. Wash clothing/bedding in hot water (>130°F / 54°C) and hot dryer cycle.



HIGH-YIELD BOARD PEARL: Permethrin must be reapplied in 9 to 10 days because it does not kill unhatched nits, allowing nymphs to emerge; re-treatment eradicates newly hatched lice before they produce new eggs.

Topic 10: Pilonidal Disease & Abscess

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Infection and abscess formation in the natal cleft of the gluteal fold, caused by hair follicles becoming obstructed, creating foreign body reaction and sinus tract.

CLINICAL PRESENTATION & PHYSICAL EXAM

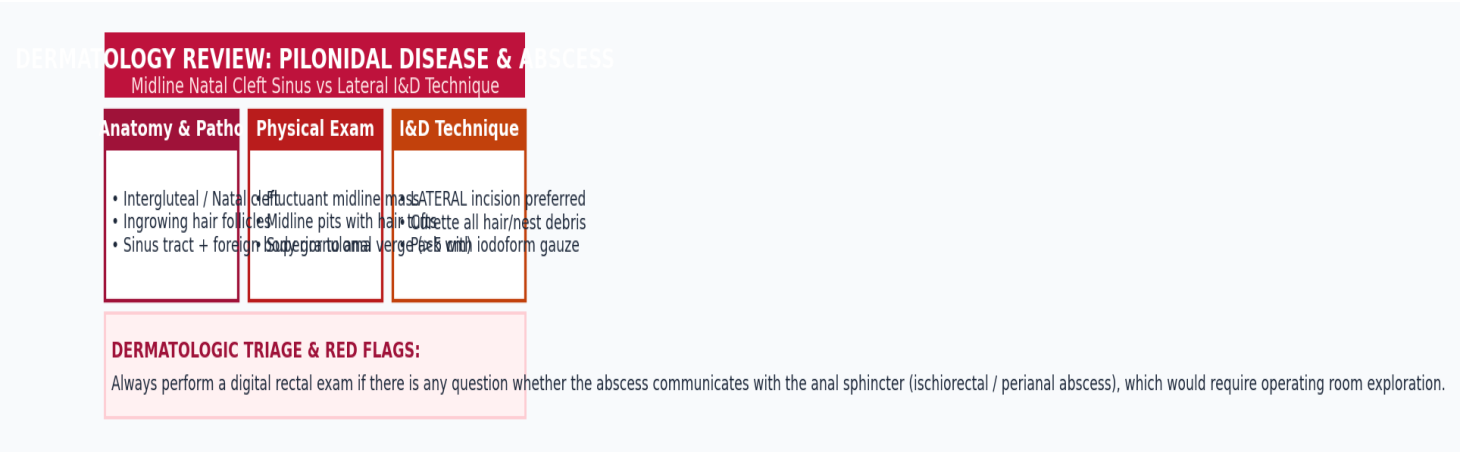
• **Presentation:** Painful, fluctuant mass in the superior gluteal cleft with drainage of purulent or bloody fluid. Exacerbated by sitting or prolonged driving.
• **Physical Exam:** Tender, erythematous, fluctuant mass in the midline sacrococcygeal region. Midline pit or sinus opening. Foul-smelling drainage.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. Differentiate from perianal abscess (closer to anal verge; requires surgical EUA).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Acute abscess: Incision and Drainage (I&D;), preferably slightly lateral to the midline cleft to promote wound healing. Thorough curettage of hair and debris, pack with iodoform gauze. Antibiotics only if extensive surrounding cellulitis or immunocompromised. Definitive surgical excision elective when uninfected.



HIGH-YIELD BOARD PEARL: Incisions for pilonidal abscess drainage should be placed slightly LATERAL to the intergluteal cleft when possible, as midline incisions heal poorly and have a much higher rate of chronic recurrence.

Topic 11: Pressure Ulcers (Staging & Wound Care)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Localized injury to the skin and underlying tissue over a bony prominence due to sustained pressure, shear, and friction, leading to microvascular occlusion and tissue ischemia.

CLINICAL PRESENTATION & PHYSICAL EXAM

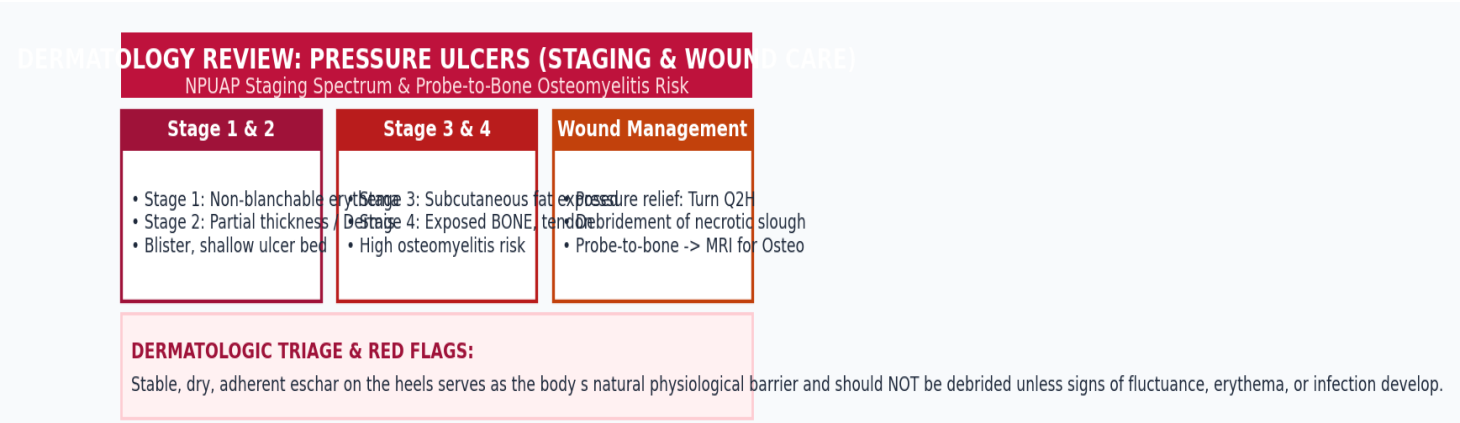
• **Presentation:** Immobile, bedridden, or elderly patient with localized erythema, ulceration, or deep necrotic tissue over sacrum, greater trochanter, or ischial tuberosities.
• **Physical Exam:** Staging: Stage 1 (Non-blanchable erythema of intact skin); Stage 2 (Partial-thickness skin loss with exposed dermis, intact/ruptured blister); Stage 3 (Full-thickness skin loss, subcutaneous fat visible); Stage 4 (Full-thickness tissue loss with exposed bone, tendon, or muscle); Unstageable (Obscured by slough or eschar).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical staging. Probe to bone test (if positive, >85% predictive of underlying osteomyelitis). Plain X-ray or MRI of underlying bone if osteomyelitis suspected.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Pressure redistribution (frequent repositioning Q2H, air-fluidized mattresses). Debridement of necrotic slough/eschar (except stable dry eschar on heels). Moisture-retentive dressings. Systemic antibiotics only if surrounding cellulitis, sepsis, or proven osteomyelitis.



HIGH-YIELD BOARD PEARL: If bone is palpable at the base of a Stage 4 pressure ulcer with a sterile metallic probe ("probe-to-bone test"), osteomyelitis is present in >85% of cases and mandates MRI imaging and targeted bone biopsy.

Topic 12: Scabies (Sarcoptes scabiei)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Infestation by the microscopic mite Sarcoptes scabiei var. hominis. Female mite burrows into stratum corneum, depositing feces and eggs, inducing Type IV hypersensitivity reaction.

CLINICAL PRESENTATION & PHYSICAL EXAM

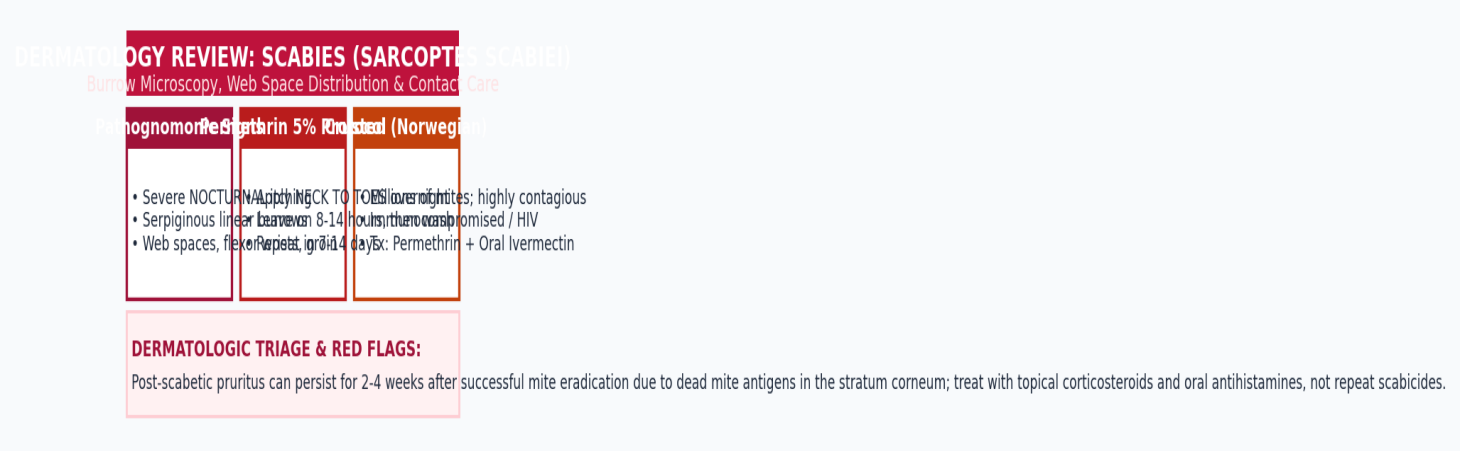
• **Presentation:** Severe, relentless, intractable nocturnal pruritus. Linear burrows in web spaces of fingers, wrists, flexor surfaces, axillae, umbilicus, and genitalia.
• **Physical Exam:** Pathognomonic serpiginous linear burrows. Small erythematous papules and vesicles. Crusted (Norwegian) scabies: thick hyperkeratotic scales teeming with millions of mites (seen in HIV/immunocompromised).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Mineral oil skin scraping of burrow showing mites, eggs, or scybala (fecal pellets) under light microscopy. Often diagnosed clinically.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

First-line: Topical Permethrin 5% cream applied from neck down to soles of feet, left on for 8-14 hours, then washed off; REPEAT IN 7 DAYS. Treat ALL household contacts and sexual partners simultaneously! Oral Ivermectin (200 mcg/kg, repeat in 14 days) for crusted scabies or institutional outbreaks. Wash bedding in hot cycle.



HIGH-YIELD BOARD PEARL: Permethrin 5% cream must be applied from the neck down to the tips of the toes and left on for 8-14 hours before washing. All close contacts and household members must be treated simultaneously, even if asymptomatic.

Topic 13: Envenomations: Brown Recluse, Black Widow & Pit Viper

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Brown Recluse (Loxosceles reclusa): Sphingomyelinase D toxin causing cytotoxic necrosis. Black Widow (Latrodectus mactans): Alpha-latrotoxin (massive acetylcholine release at motor endplates). Pit Viper (Crotalinae: rattlesnakes): metalloproteinases, hemotoxins causing coagulopathy and tissue destruction.

CLINICAL PRESENTATION & PHYSICAL EXAM

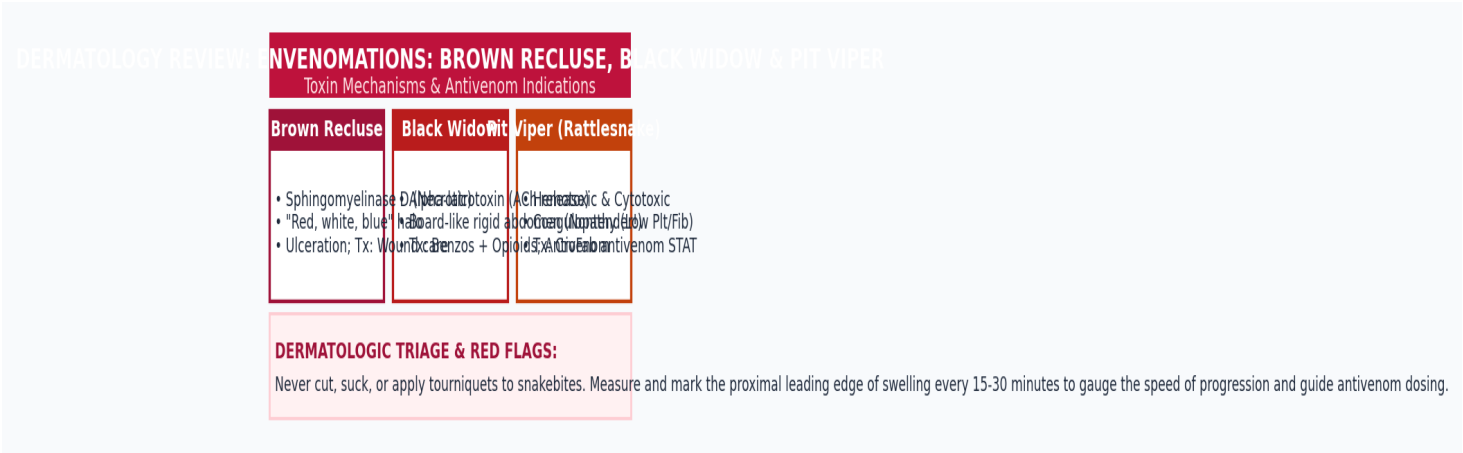
• **Presentation:** Recluse: painless bite -> "red, white, and blue" halo -> central necrotic ulcer. Widow: pinprick bite -> severe muscle spasms, rigid board-like abdomen without tenderness, facial grimace (facies latrodectismica). Pit Viper: intense local swelling, ecchymosis, bullae, coagulopathy.
• **Physical Exam:** Recluse: central violaceous depression with surrounding ischemic pallor and erythema. Widow: target bite marks, muscle rigidity, hypertension. Pit viper: fang marks, swelling progressing across joints, bleeding.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical. Pit viper: STAT Coagulation panel (PT/INR, Fibrinogen, Platelets).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Recluse: wound care, elevation, tetanus; avoid early excision. Widow: IV Opioids + Benzodiazepines (Diazepam) for spasms; Antivenom (Equine) for refractory life-threatening spasms/HTN. Pit Viper: CroFab antivenom (indicated for progressing swelling, coagulopathy [platelets<100k, fibrinogen<100], or systemic shock). NEVER apply tourniquets, ice, or suction.



HIGH-YIELD BOARD PEARL: Black Widow bites cause severe systemic muscle rigidity and painful abdominal cramping ("board-like abdomen") that mimics an acute surgical abdomen, but the abdomen is nontender to palpation. Pit Viper envenomation with progressive swelling or coagulopathy requires CroFab antivenom.

Topic 14: Stevens-Johnson Syndrome (SJS) Spectrum

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Severe immune-mediated mucocutaneous reaction characterized by extensive keratinocyte apoptosis. SJS: < 10% TBSA epidermal detachment. SJS/TEN overlap: 10-30% TBSA. High-risk drugs: SATAN (Sulfonamides, Allopurinol, Anticonvulsants [carbamazepine, lamotrigine, phenytoin], NSAIDs/Oxicams, Nevirapine).

CLINICAL PRESENTATION & PHYSICAL EXAM

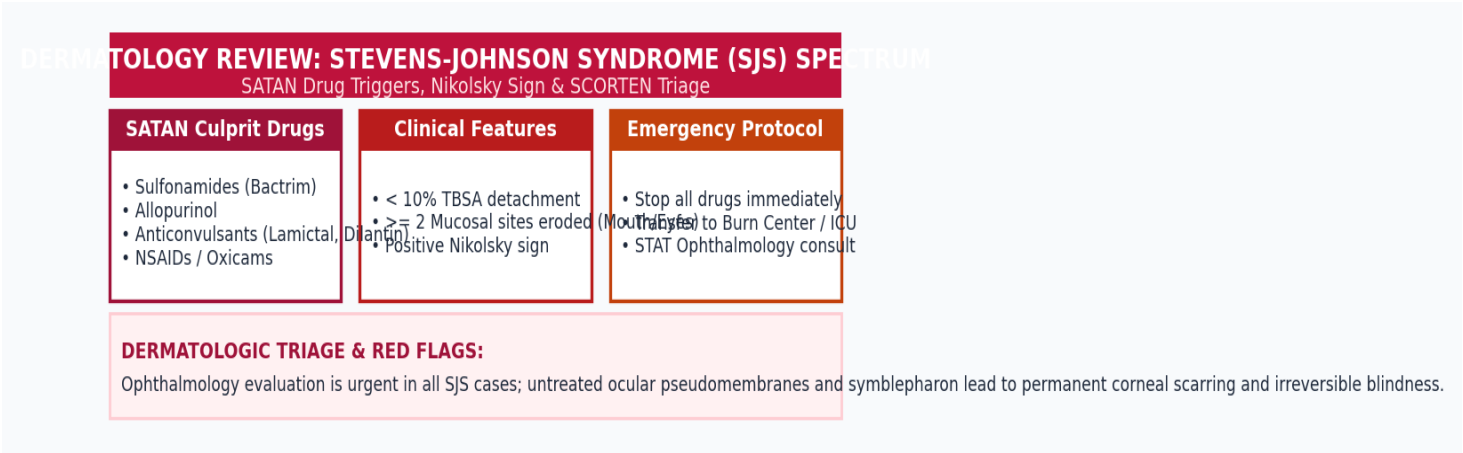
• **Presentation:** Fever and flu-like prodrome 1-3 weeks after drug initiation. Painful targetoid lesions spreading rapidly, followed by epidermal blistering and sheet-like sloughing. Severe painful oral, conjunctival, and genital ulcerations.
• **Physical Exam:** Atypical target lesions, flaccid bullae, positive Nikolsky sign (sloughing with lateral pressure). Mucosal erosion in at least 2 distinct mucosal sites (hallmark).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis confirmed by frozen-section skin biopsy showing full-thickness epidermal necrosis with minimal dermal inflammation. Calculate SCORTEN score for mortality risk.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

IMMEDIATE DISCONTINUATION of all potential culprit drugs. STAT transfer to a specialized Burn Center or ICU. Aggressive fluid resuscitation (guided by urine output), strict reverse isolation, sterile wound care. STAT Ophthalmology consult to prevent cicatricial blindness. IVIG or Cyclosporine.



HIGH-YIELD BOARD PEARL: SJS involves < 10% TBSA epidermal detachment and mandates mucosal involvement in at least TWO distinct sites (oral, conjunctival, or urogenital). Prompt transfer to a Burn ICU significantly reduces mortality.

Topic 15: Toxic Epidermal Necrolysis (TEN)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Life-threatening mucocutaneous reaction representing the maximum severity of epidermal necrolysis with > 30% TBSA epidermal detachment. Mediated by granulysin and Fas-Fas ligand triggering massive keratinocyte apoptosis.

CLINICAL PRESENTATION & PHYSICAL EXAM

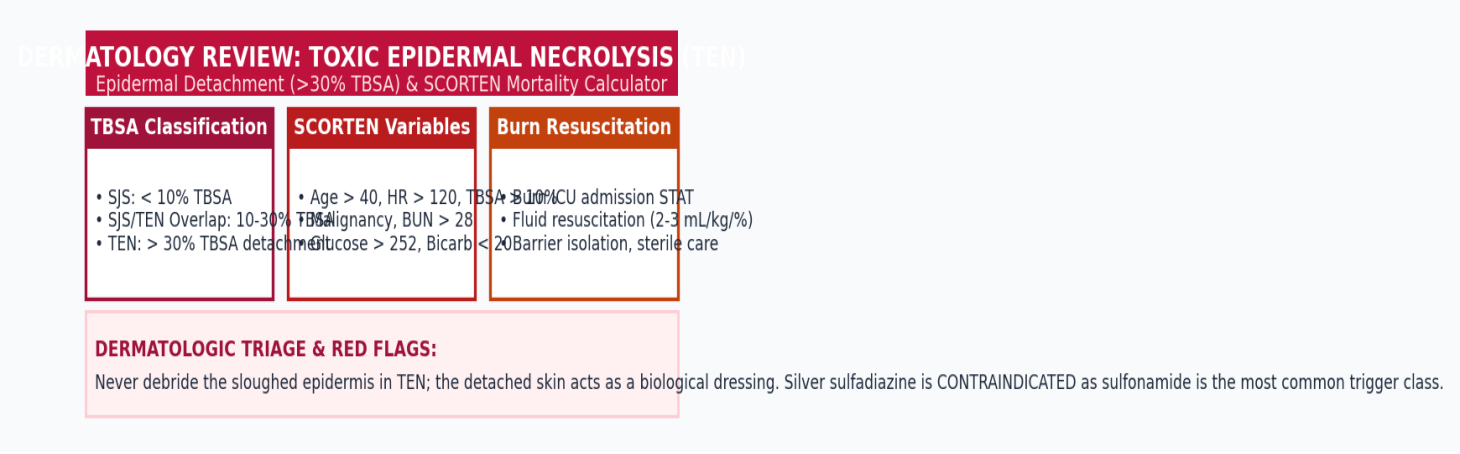
• **Presentation:** High fever, severe cutaneous pain, diffuse erythema, and flaccid blisters that rapidly coalesce, resulting in widespread, full-thickness sheet-like shedding of skin ("scalded" appearance). Hemorrhagic mucosal crusting.
• **Physical Exam:** Positive Nikolsky sign, positive Asboe-Hansen sign. Raw, denuded, weeping dermis resembling extensive second-degree thermal burns. Severe purulent conjunctivitis, tracheobronchial sloughing causing respiratory distress.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Skin biopsy (full-thickness epidermal necrosis). SCORTEN assessment (Age >40, Malignancy, TBSA >10%, Heart rate >120, BUN >28, Glucose >252, Bicarb <20; each adds 1 point).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Burn ICU admission. Immediate cessation of all non-essential drugs. Fluid and electrolyte replacement with Lactated Ringer (lower volume than thermal burns). Nutritional support, temperature regulation. Sterile non-adherent dressings. Early IVIG, Cyclosporine, or Etanercept.



HIGH-YIELD BOARD PEARL: Toxic Epidermal Necrolysis involves > 30% TBSA epidermal detachment and carries a mortality rate of 30-50%. SCORTEN calculated within the first 24 hours objectively predicts mortality and guides intensive resuscitation.

Topic 16: Acute Urticaria & Anaphylaxis Triage

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Mast cell degranulation with histamine and leukotriene release into superficial dermis (urticaria) or deep dermis/subcutis (angioedema). IgE-mediated Type I hypersensitivity (foods, drugs, venom) or non-IgE (radiocontrast, NSAIDs, viral).

CLINICAL PRESENTATION & PHYSICAL EXAM

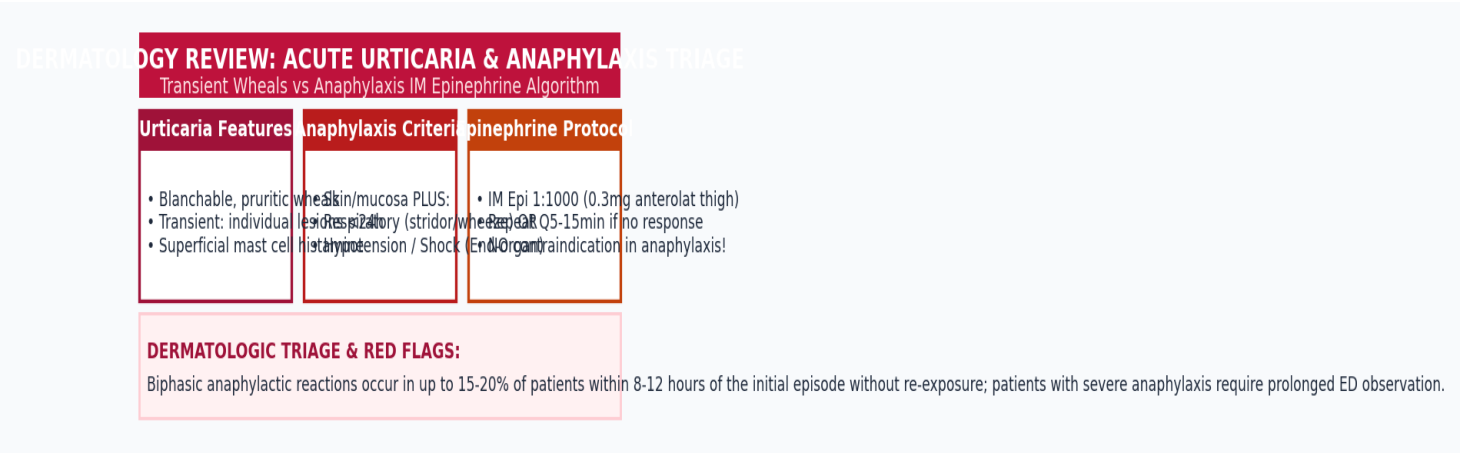
• **Presentation:** Urticaria: intensely pruritic, transient, raised, blanchable wheals that resolve within 24 hours without scarring. Anaphylaxis: acute onset involving skin/mucosa PLUS respiratory compromise (stridor, wheezing) or hypotension.
• **Physical Exam:** Circumscribed edematous papules and plaques with central pallor and erythematous halos. Stridor, wheezing, uvular edema, tongue swelling indicate life-threatening airway compromise.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical. Anaphylaxis criteria: 1) Acute onset of skin/mucosal symptoms PLUS respiratory distress OR hypotension; OR 2) Two or more organ systems involved after known exposure.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Isolated Urticaria: Second-generation oral H1 antihistamine (Cetirizine 10-20 mg daily) +/- H2 blocker (Famotidine 20 mg BID). Short course Prednisone for severe refractory cases. Anaphylaxis: INTRAMUSCULAR EPINEPHRINE 1:1,000 (0.3 mg IM in anterolateral thigh for adults; 0.15 mg for pediatrics) GIVEN IMMEDIATELY. Repeat Q5-15 min PRN. IV crystalloid boluses for shock.



HIGH-YIELD BOARD PEARL: Intramuscular Epinephrine (0.3 mg 1:1,000 in anterolateral thigh) is the definitive first-line medication for anaphylaxis; there are NO absolute contraindications to epinephrine in anaphylaxis. Antihistamines and steroids are secondary adjuncts.

Topic 17: Pediatric Viral Exanthems

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Classic childhood exanthems: Measles (Rubeola: Paramyxovirus), Rubella (German Measles: Togavirus), Erythema Infectiosum (Fifth Disease: Parvovirus B19), Roseola Infantum (Sixth Disease: HHV-6), Varicella (VZV), Hand-Foot-and-Mouth (Coxsackievirus A16).

CLINICAL PRESENTATION & PHYSICAL EXAM

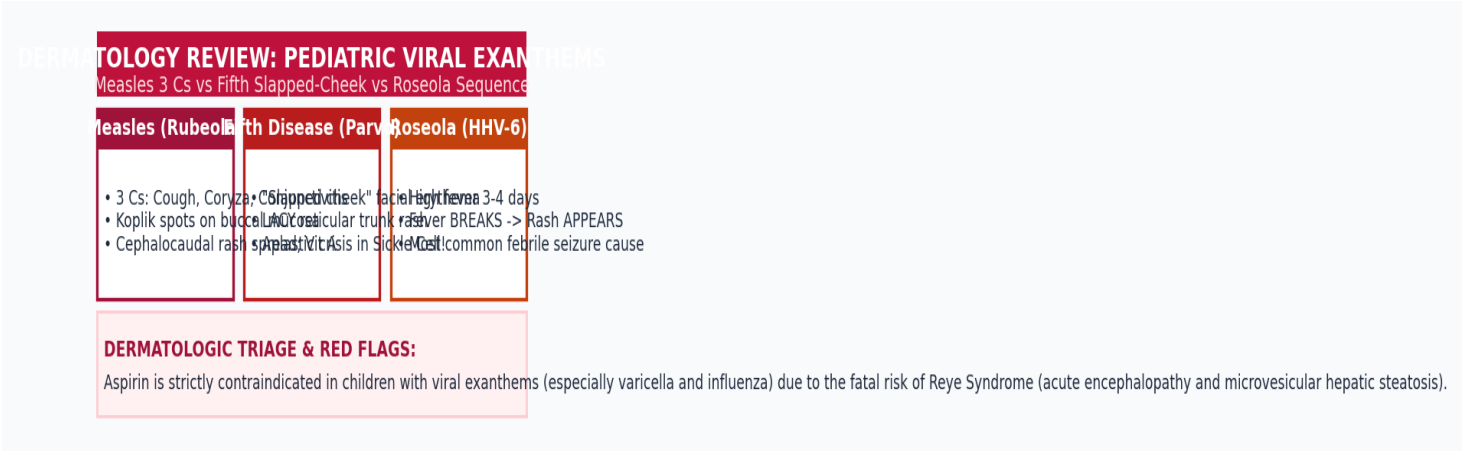
• **Presentation:** Measles: Cough, Coryza, Conjunctivitis (3 Cs) + Koplik spots -> cephalocaudal rash. Fifth: "slapped-cheek" rash -> reticular lace-like rash on trunk/limbs. Roseola: High fever for 3-4 days that resolves abruptly, followed immediately by pink maculopapular rash on trunk.
• **Physical Exam:** Koplik spots: tiny white papules on buccal mucosa opposite molars. Slapped-cheek facial erythema with circumoral pallor. Hand-Foot-Mouth: painful oval vesicles on palms, soles, and oral mucosa.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical. Parvovirus B19 serology (IgM) if pregnant contact exposed (risk of hydrops fetalis).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Supportive care: hydration, antipyretics (Acetaminophen, Ibuprofen; AVOID ASPIRIN due to Reye Syndrome). Measles: Vitamin A supplementation reduces morbidity/mortality. Isolate suspected measles (airborne precautions).



HIGH-YIELD BOARD PEARL: Roseola (HHV-6) classically presents with high fever lasting 3 to 5 days; the fever abruptly disappears, and THEN a pink maculopapular rash appears on the trunk. Parvovirus B19 can trigger transient aplastic crisis in patients with sickle cell disease and hydrops fetalis in pregnancy.